

The secondary magnetic field depends on the external magnetic field

$$\mathbf{B}_{\text{sec}} = \sigma \mathbf{B}_0$$

σ is called the shielding constant (diamagnetic shielding constant)

$$\mathbf{B}_{\text{eff}} = \mathbf{B}_0 - \mathbf{B}_{\text{sec}}$$

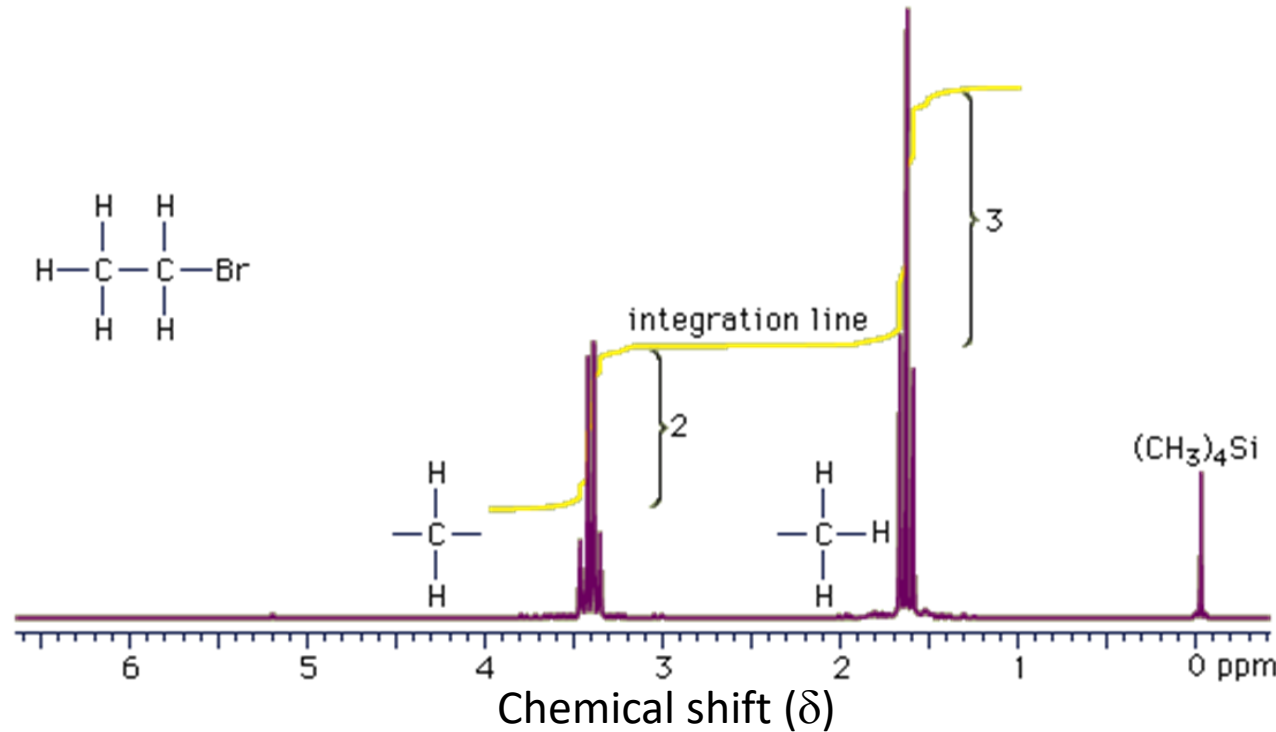
$$\mathbf{B}_{\text{eff}} = \mathbf{B}_0 (1 - \sigma)$$

- The shielding constant determines the resonance frequency
- These different frequency values are termed as **chemical shifts**

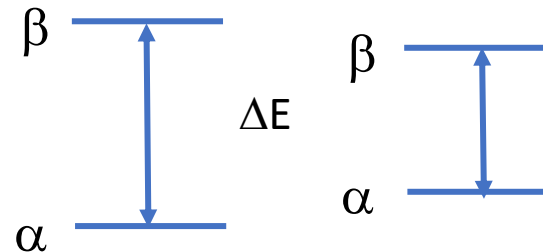
Shielding and Deshielding

$$B_{\text{eff}} = B_0 - B_{\text{local}}$$

$$\Delta E = h(\gamma/2\pi)B_{\text{eff}}$$



e poor environment
 Deshielded proton
 Down field
 Higher frequency
 Larger δ



e rich environment
 Shielded proton
 Up field
 Lower frequency
 Lower δ

Greater electron density --- greater B_{local} --- smaller B_{eff} --- smaller ΔE --- lower frequency --- lower chemical shift

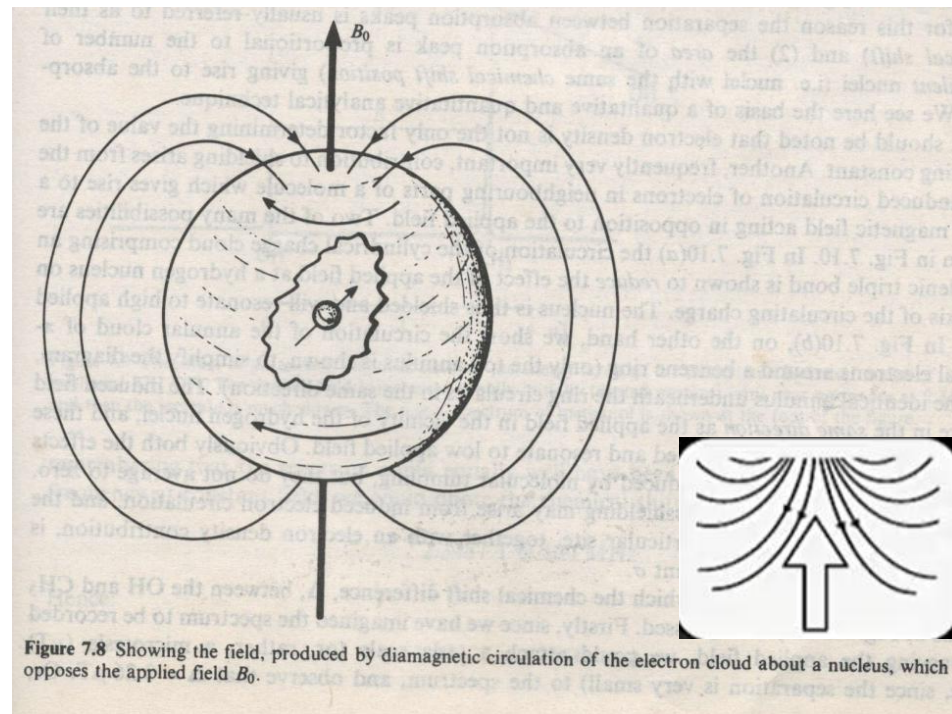
Shielding and deshielding

At a particular magnetic field strength, we might expect all ^1H nuclei to absorb the same frequency of the electromagnetic radiation (RF). But this is not true because the nuclei are surrounded by electrons and these electrons shield the nucleus. In the presence of an external magnetic field, the electrons circulate which generate a local (induced) magnetic field. The direction of this induced magnetic field is opposite to that of the external magnetic field. This effect is called diamagnetic shielding. More the electron density around the nucleus, more is the shielding of the nucleus. This effect is particularly important for NMR spectroscopy. Without this effect, all protons would absorb the same frequency and hence would not provide any useful information about the molecule.

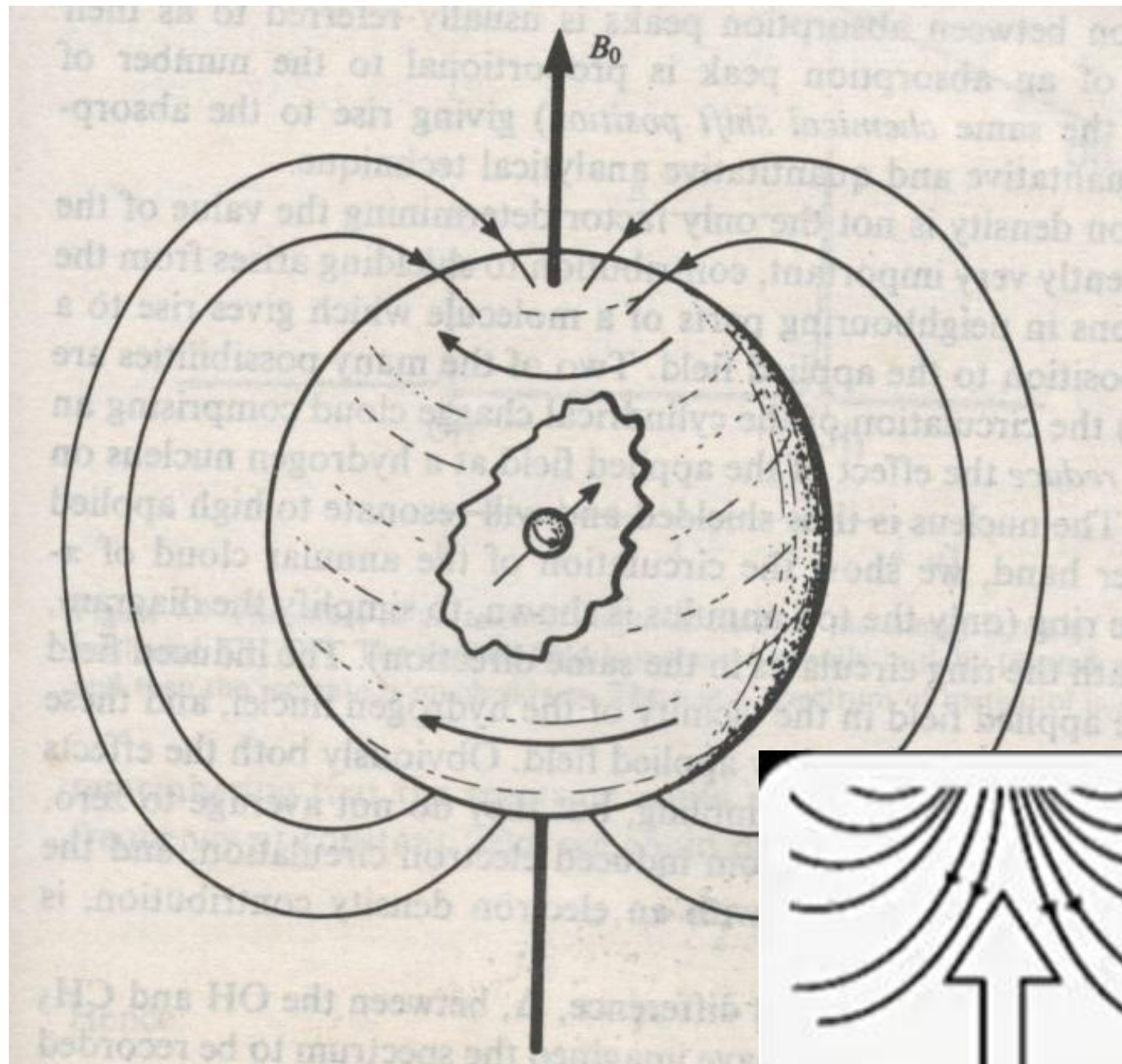
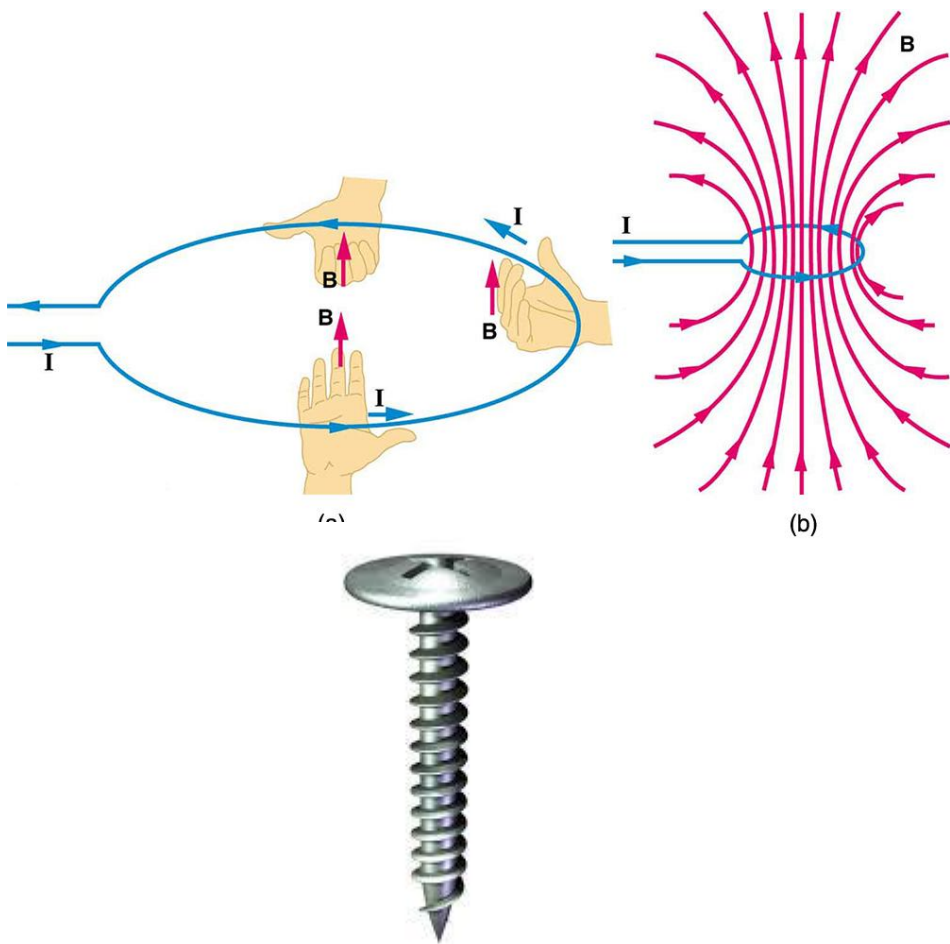
The magnitude of induced magnetic field is small but has important effect on the nucleus. Now the proton experiences two magnetic fields i.e. the strong, external magnetic field and the weak induced magnetic field produced by the circulating electron density. Hence the net/effective magnetic field (B_{eff}) felt by the proton is slightly smaller than the external magnetic field and the proton is said to be shielded by the electrons.

$$B_{\text{eff}} = B_0 - B_{\text{local}}$$

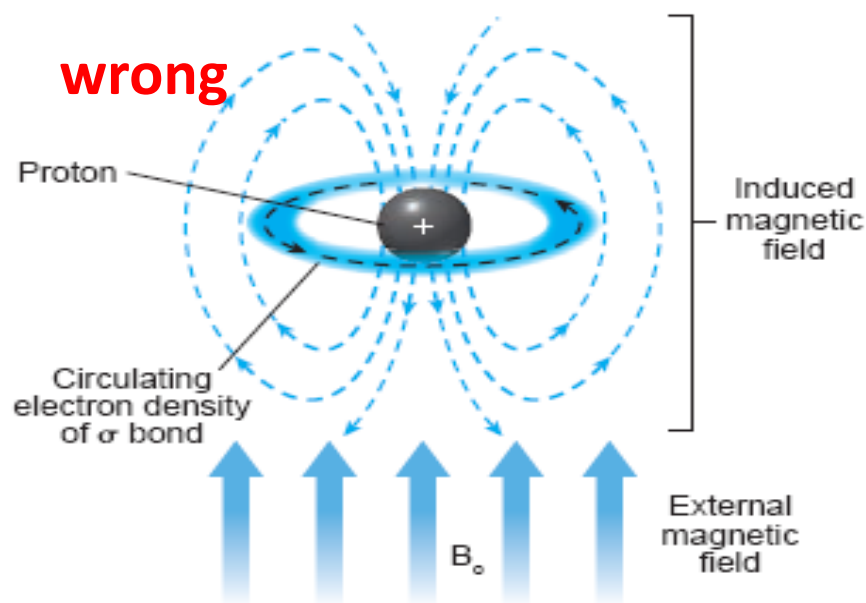
They therefore require a *lower frequency* to come into resonance--that is flip their spin because ΔE is smaller....



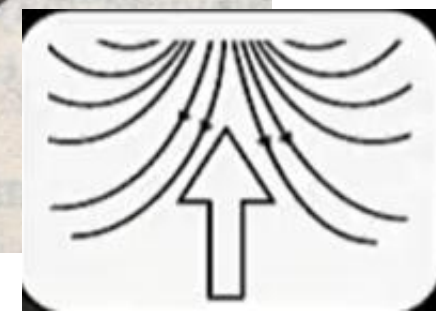
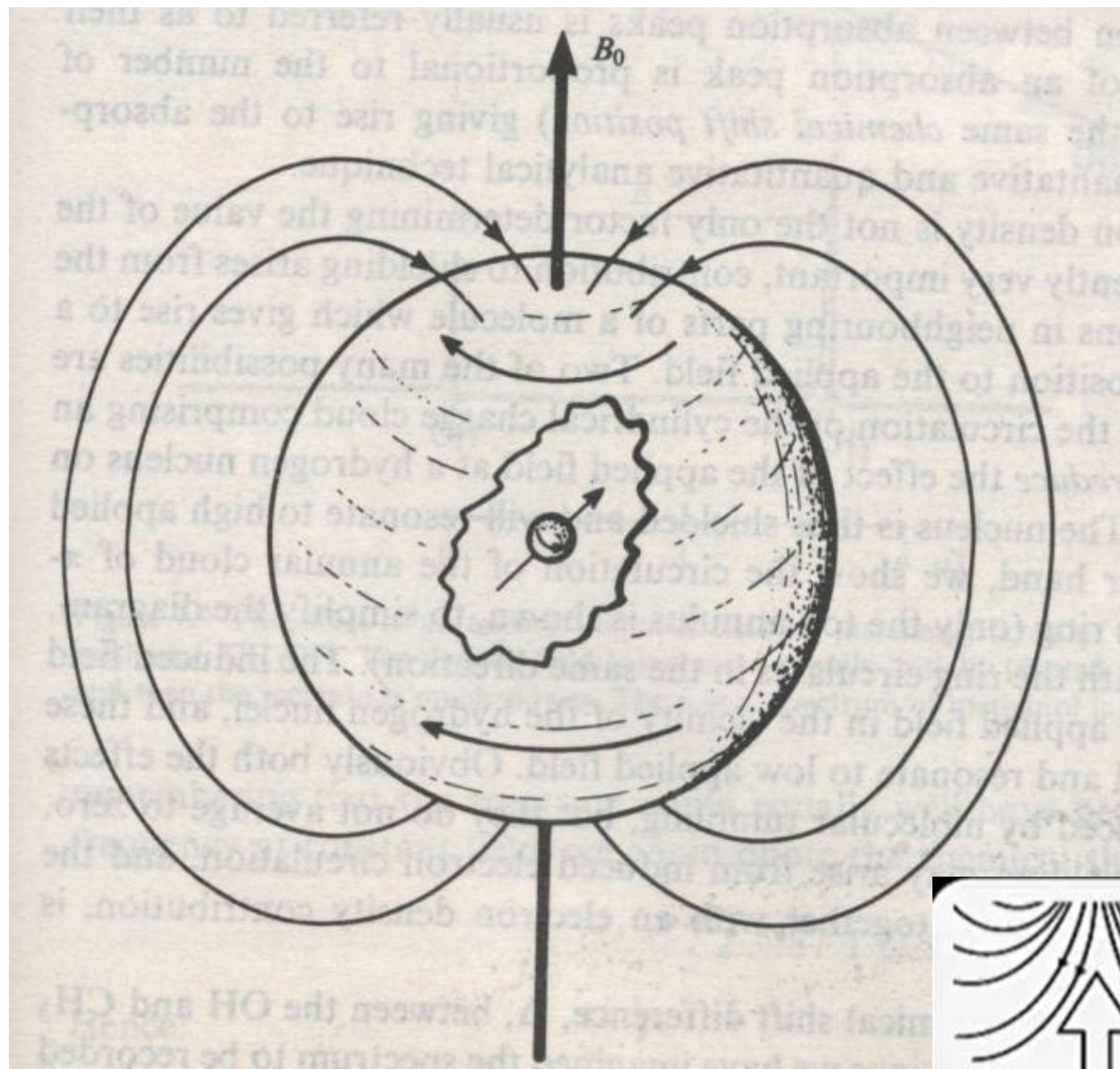
when placed in a magnetic field, the surrounding electron cloud tends to circulate in such a direction as to oppose as to produce a field opposing that applied mag field as shown in the figure below. (so called diamagnetic circulation)



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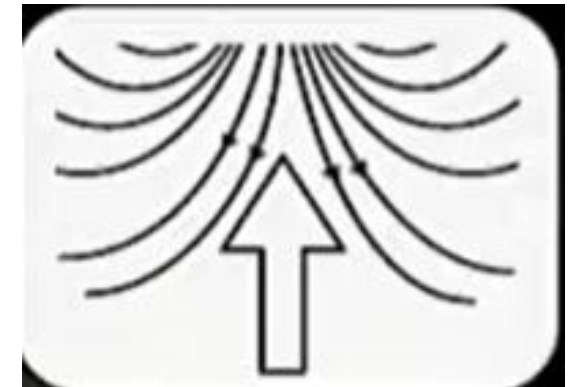
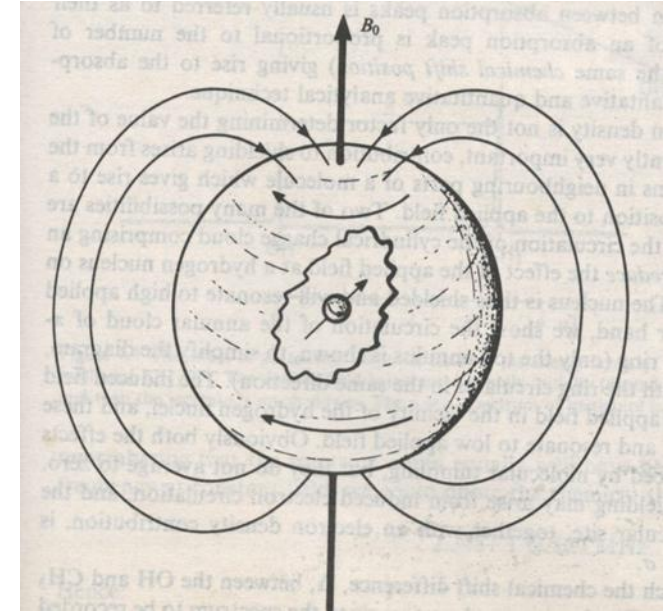


The induced magnetic field generated by the motion of electrons that are surrounding the proton.



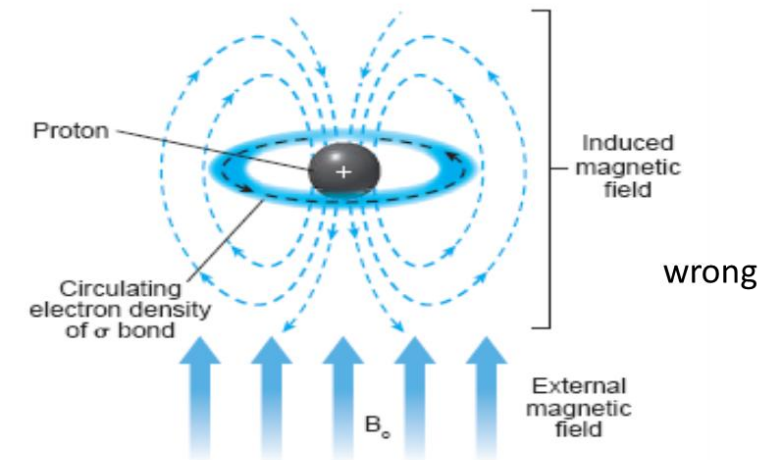
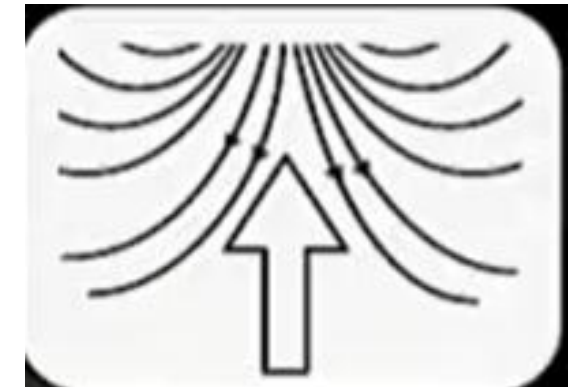
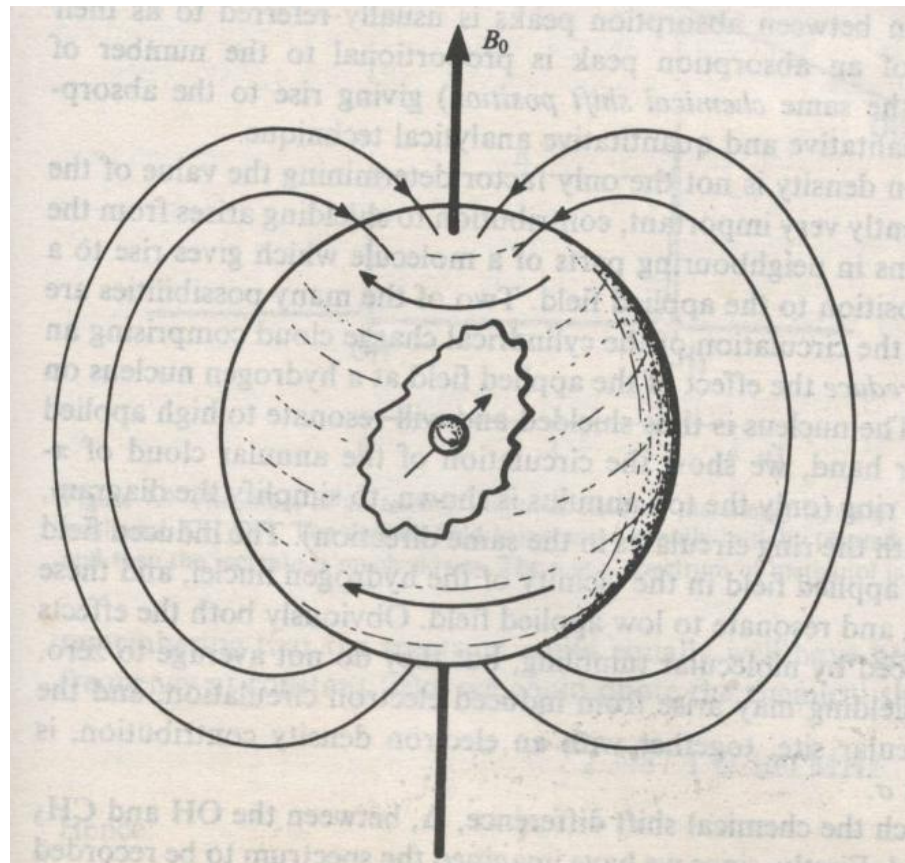
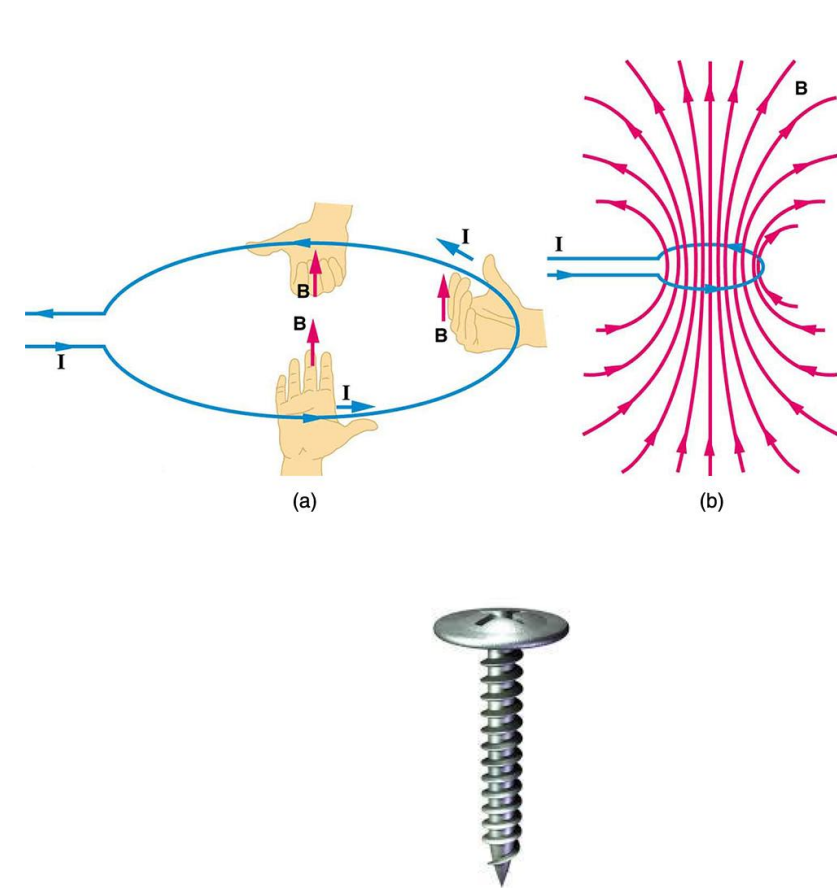
Shielding and Deshielding

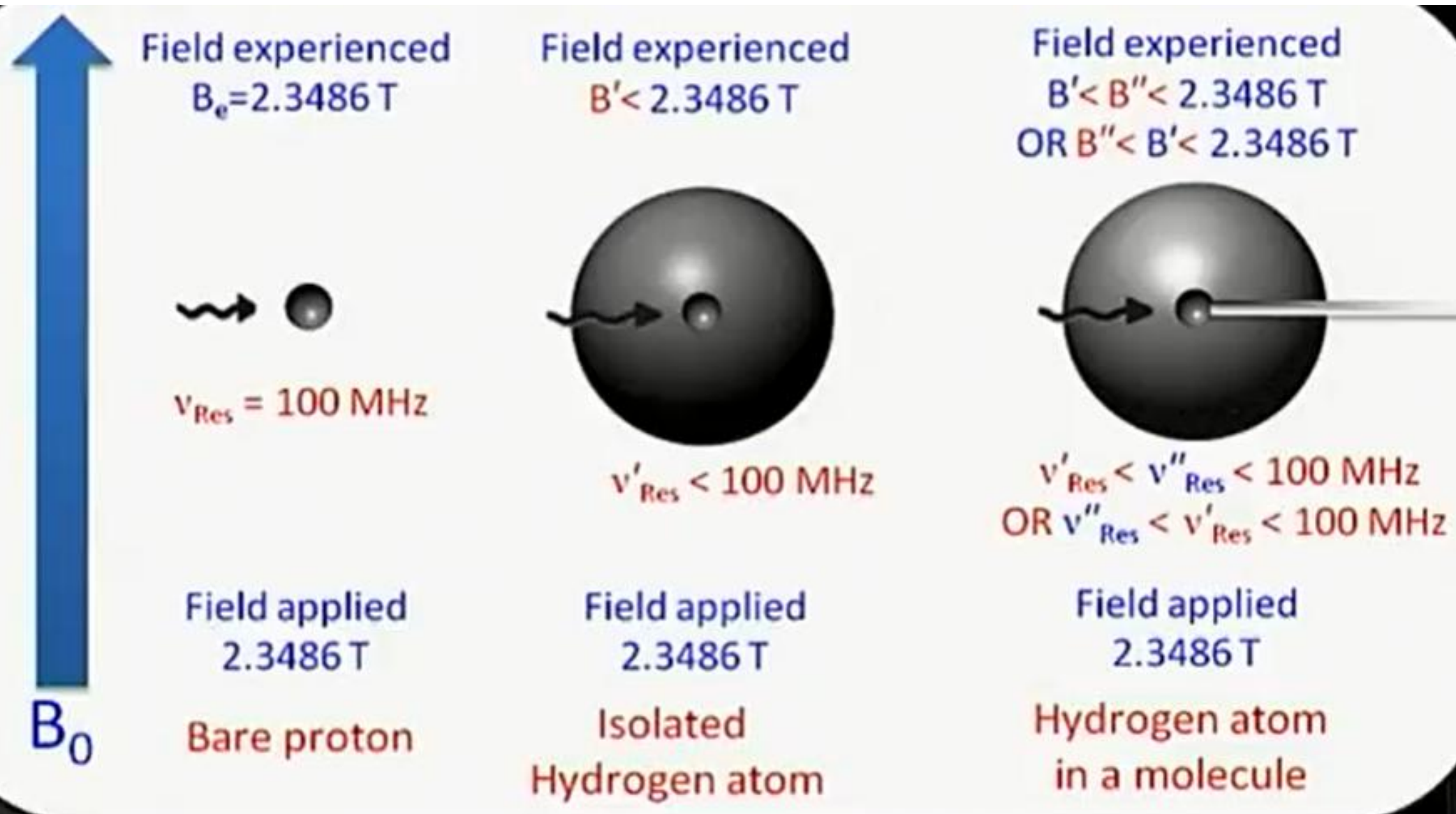
- ^1H nuclei are shielded by surrounding electrons, which generate an induced magnetic field (B_{local}) opposing the external field B_0 (diamagnetic shielding).
- More the electron density around the nucleus, more is the shielding of the nucleus, less is the effective magnetic field ($B_{\text{eff}} = B_0 - B_{\text{local}}$) experienced by the nucleus.
- Shielded protons require a lower frequency to resonate due to a smaller energy difference ($\Delta E = h(\gamma/2\pi)B_{\text{eff}}$).
- Some protons are surrounded by more electron density and are more shielded, while other protons are surrounded by less electron density and are less shielded or deshielded. (paramagnetic deshielding)
- Shielding and deshielding effects differentiate proton signals, making NMR spectroscopy informative for molecular analysis.



Shielding and Deshielding

When placed in a magnetic field, the surrounding electron cloud tends to circulate in such a direction as to oppose as to produce a field opposing that applied mag field as shown in the figure below. (so called diamagnetic circulation)





Nuclear Shielding

The electrons circulating around the nucleus produce an induced magnetic field (B_{ind})

It opposes the applied field (B_o)

The effect is called diamagnetic effect

The magnetic field experienced by the proton is less than the applied field

Mathematically;

If the applied field be B_o and the induced field B_{ind}

Effective field experienced by the nuclei, $B_{\text{eff}} = B_o - B_{\text{ind}}$

$$\gamma = \frac{e}{2m_p} g = g\mu_N/\hbar,$$

The induced field is proportional to the applied field by $B_{\text{ind}} = -\sigma B_o$

(σ is called the shielding constant and is a measure of the extent of shielding)

Substituting the value of B_{ind} we get,

$$B_{\text{eff}} = B_o - \sigma B_o$$

$$B_{\text{eff}} = (1 - \sigma) B_o$$

Due to the extra nuclear electrons the resonance conditions get modified $\nu = g_N \mu_N (1 - \sigma) B_o / h = \gamma (1 - \sigma) B_o / 2\pi$

In order to get the resonance at the fixed frequency, the applied field needs to be increased (upfield)

In order to get the resonance at the fixed applied field, the frequency needs to be decreased (upfield)

A few important points:

- To observe an NMR signal, the applied radiofrequency should match the Larmor frequency: *resonance*
- Energy difference between the spin-states is typically quite small: small population difference
- The signal saturates quickly; multiple scans are required
- Population difference increases at higher magnetic fields: less number of scans
- In a molecule, the magnetic field experienced by a group of protons (nuclei) is influenced by electrons in the nearby bonds (*chemical shifts*) as well as protons from other groups (*spin-spin coupling*): Keys to structural elucidation by NMR

The Unit of Chemical Shift

- Different kinds of NMR spectrometers operate under a range of magnetic fields (1.4-21 T)

ω at 1.4 T: 60 MHz

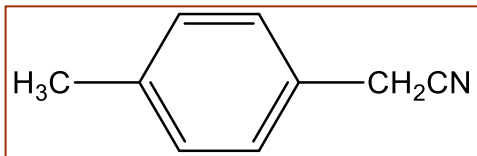
ω at 21 T: 900 MHz

- Chemical shifts given in Hz will vary from one instrument to other
- It is important to develop a scale to express chemical shifts in a form which will be independent of magnetic field

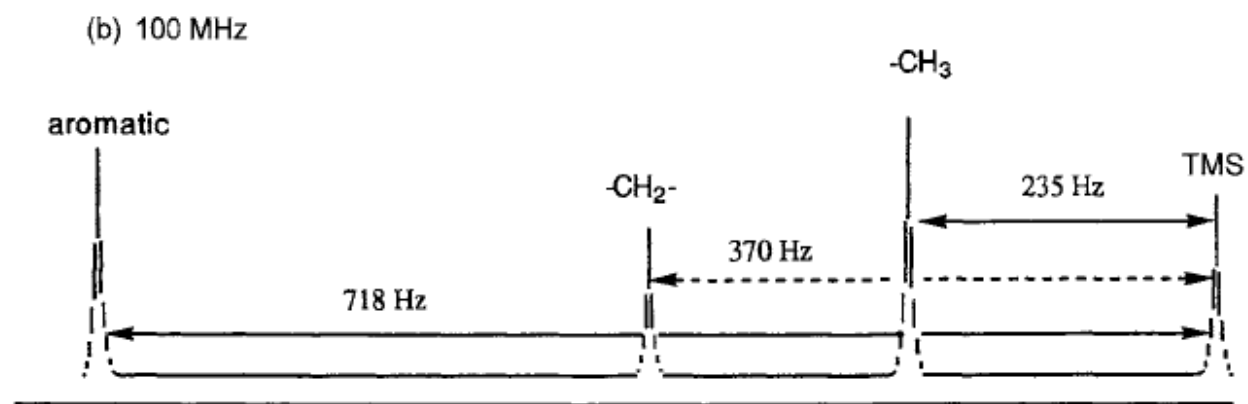
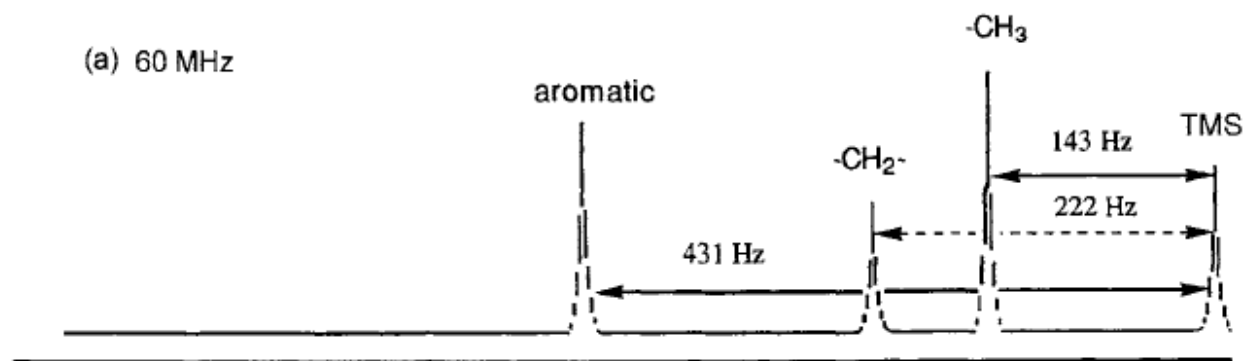
$$\Delta\nu = \nu_{\text{sample}} - \nu_{\text{standard(TMS)}}$$

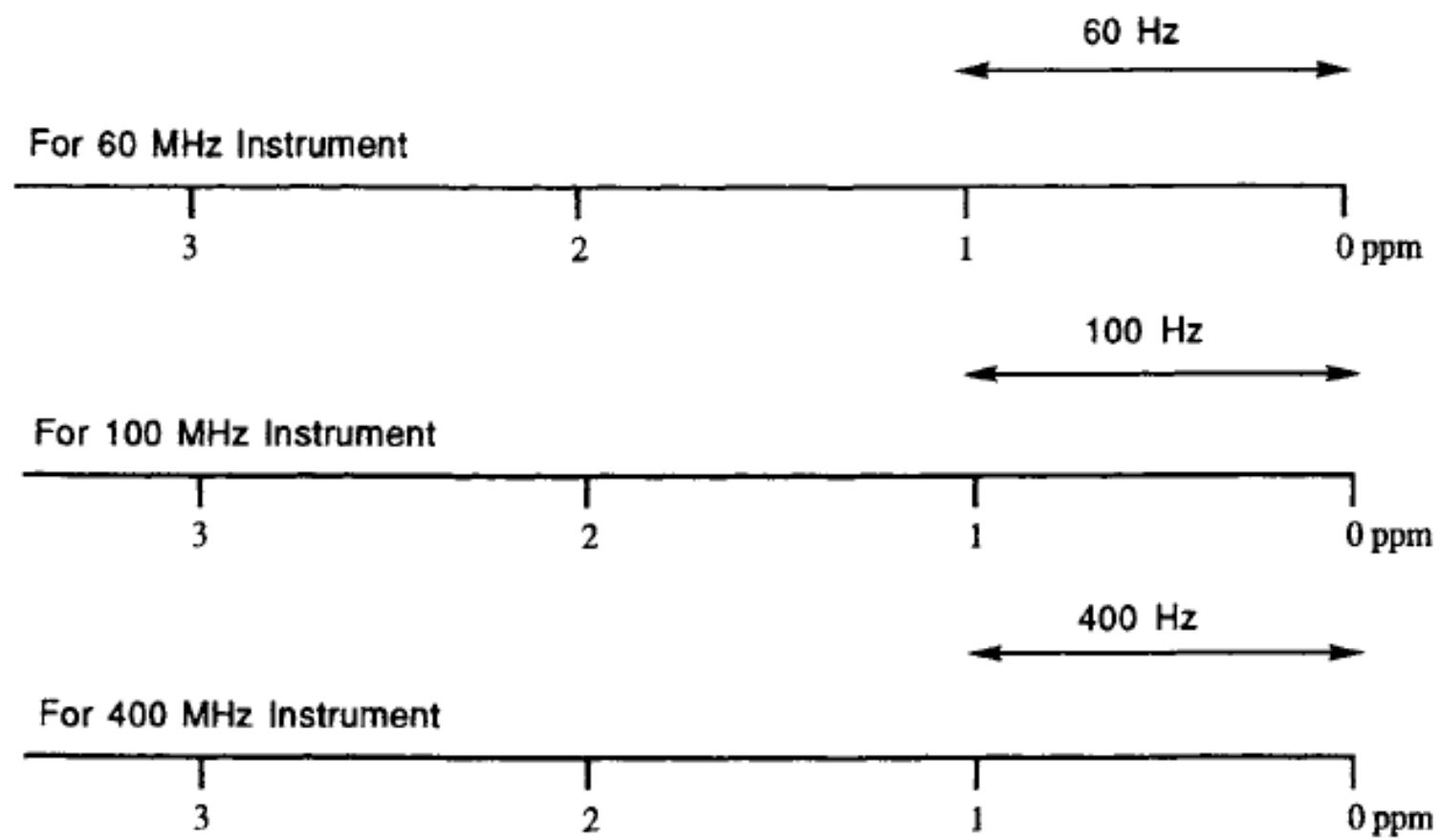
$$\delta = \frac{\nu_{\text{sample}} - \nu_{\text{standard}}}{\nu_{\text{spectrometer}}} \times 10^6$$

Chemical shifts are expressed in δ in ppm (parts per million)

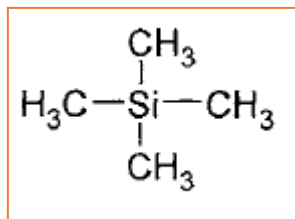


$$\delta = \frac{\Delta\nu}{\nu_{\text{spectrometer}}} \times 10^6$$

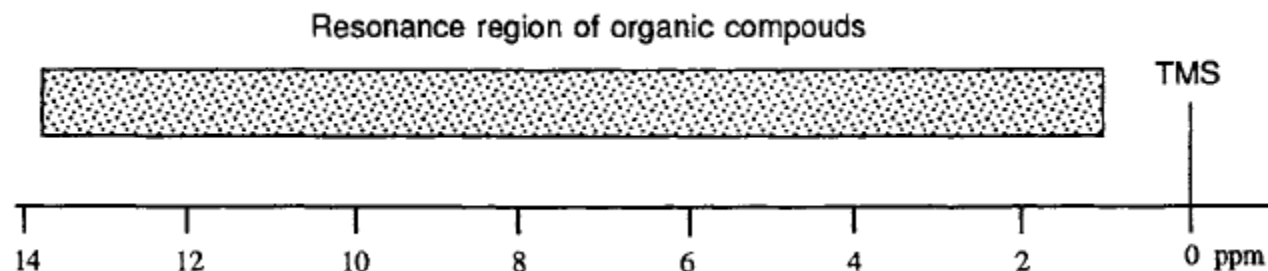




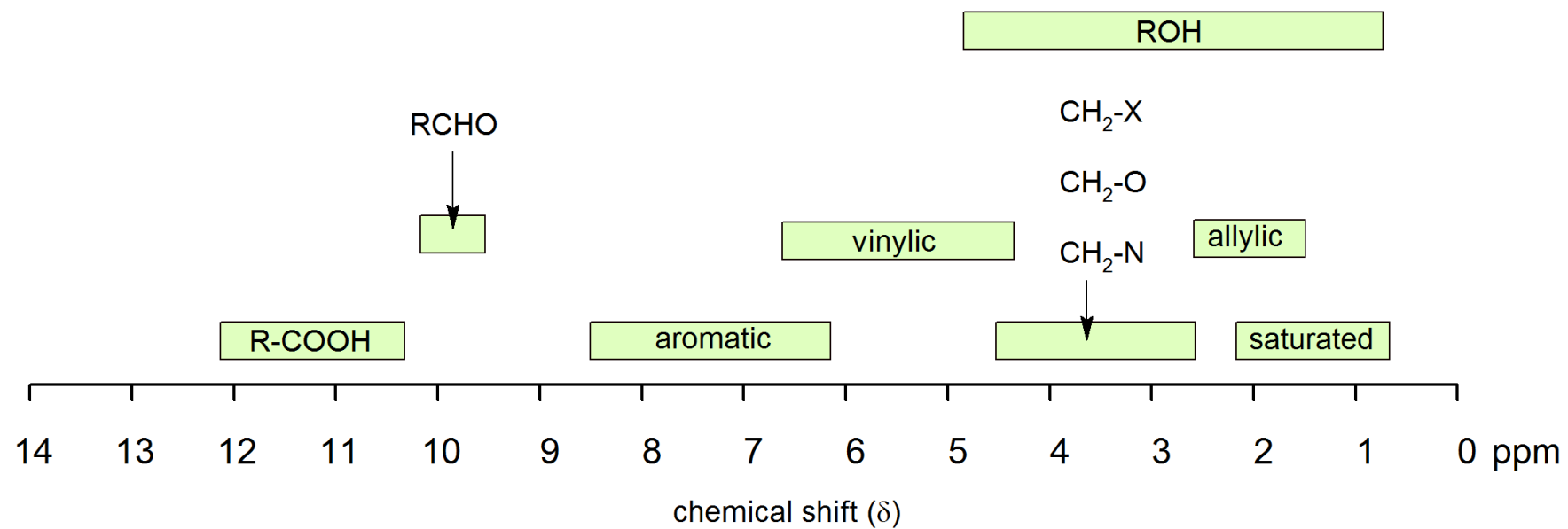
- The chemical shifts are measured with respect to protons from reference compounds
- The most generally used reference compound is tetramethyl silane (TMS)

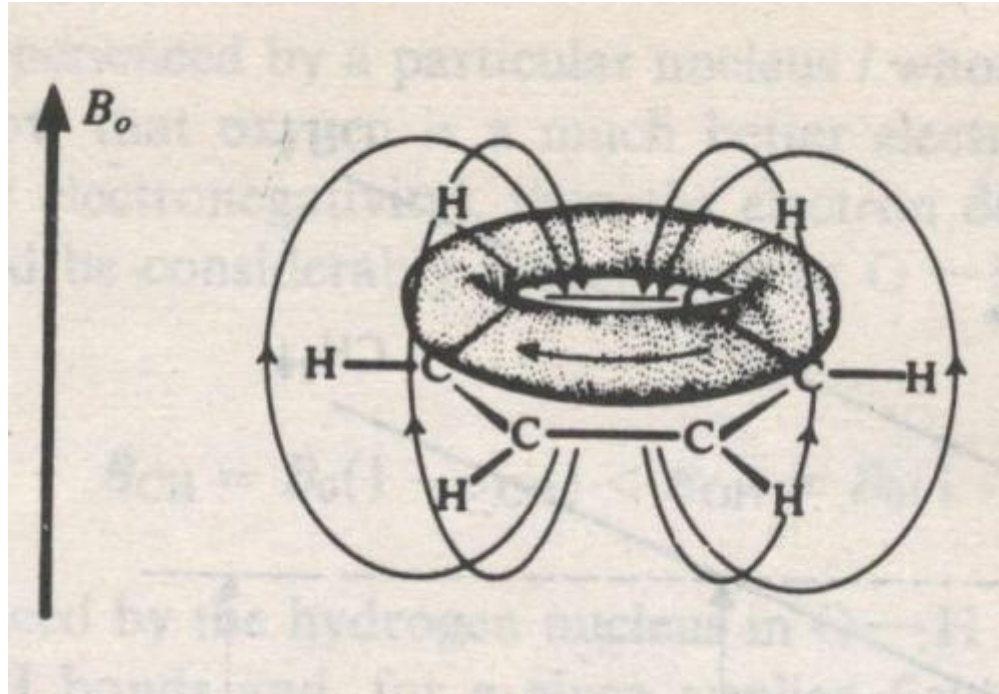


- The TMS signal appears on the right side and is generally distinguishable from most of the other resonances
- TMS is cheap, chemically inert and volatile (b.p. 27 °C)



Chemical Shifts of Organic Compounds

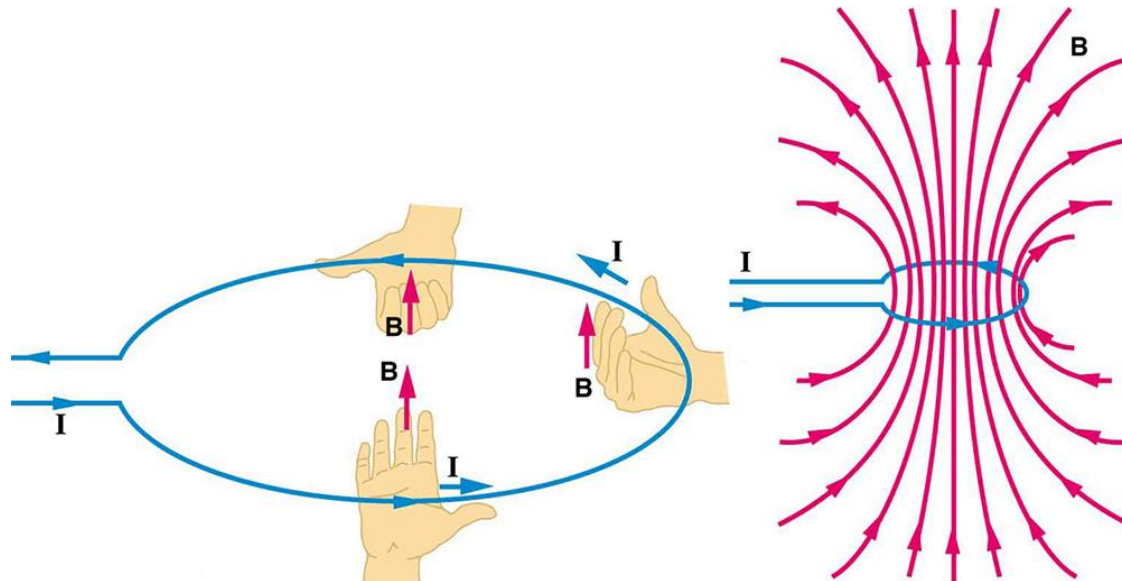




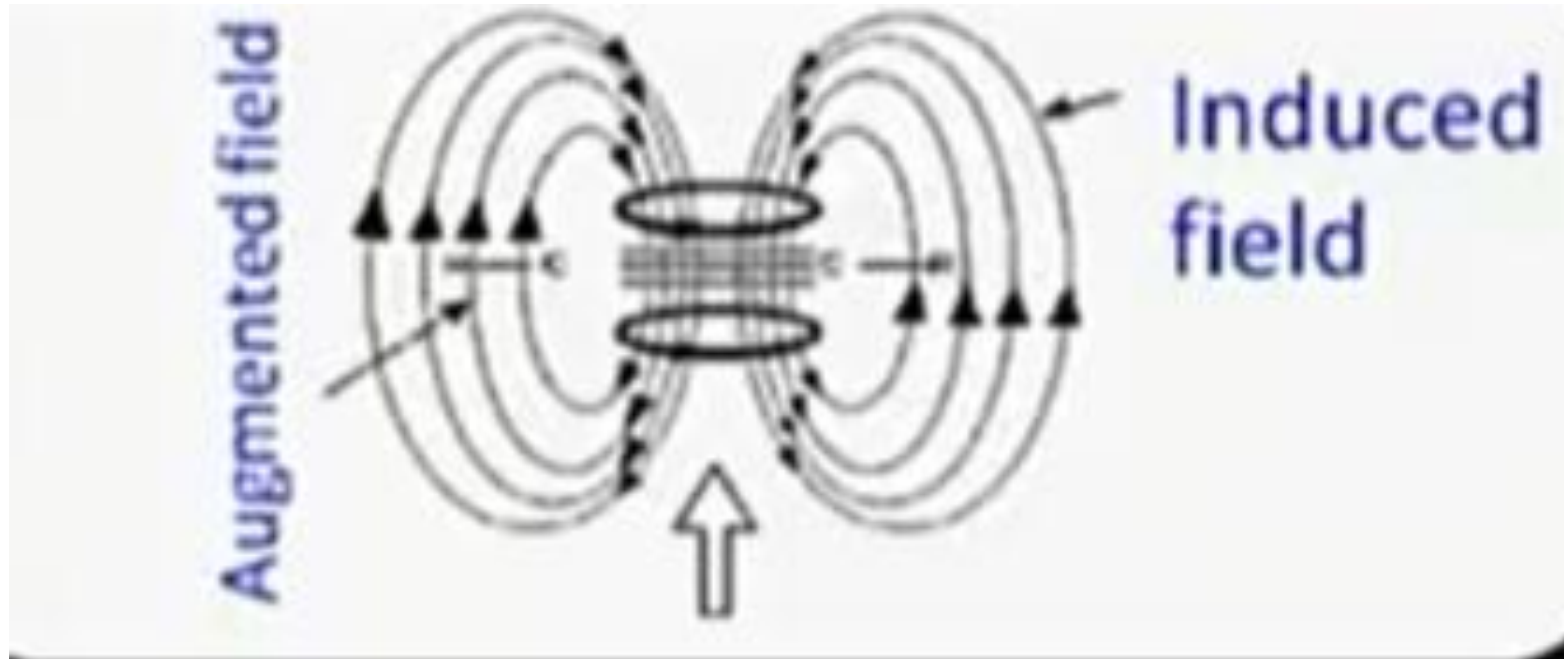
Banwell

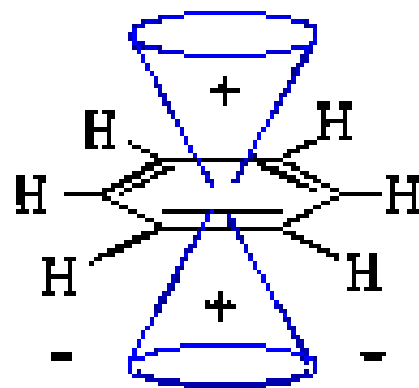


RHR
Screw (screw driver)

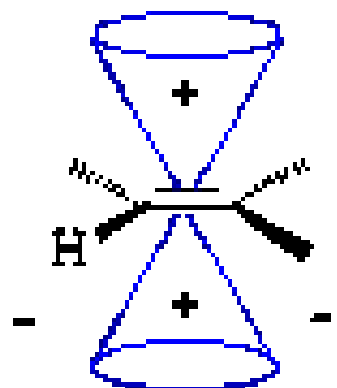


Not all protons occupy identical electronic environments. Some protons are surrounded by more electron density and are more shielded, while other protons are surrounded by less electron density and are less shielded or deshielded. (*paramagnetic deshielding*) As a result, protons in different electronic environments will exhibit a different energy gap (ΔE) between α and β spin states and will therefore absorb different frequencies of RF radiation. This phenomenon of shielding and deshielding allows in exploring the chemical environment of each hydrogen atom in a molecule.

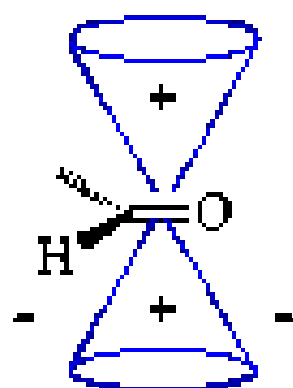




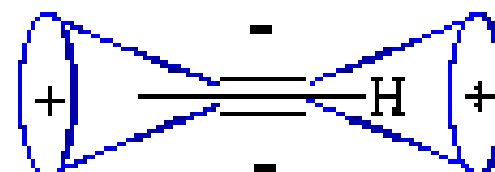
$\delta = 7-8 \text{ ppm}$



$\delta = 5-7 \text{ ppm}$



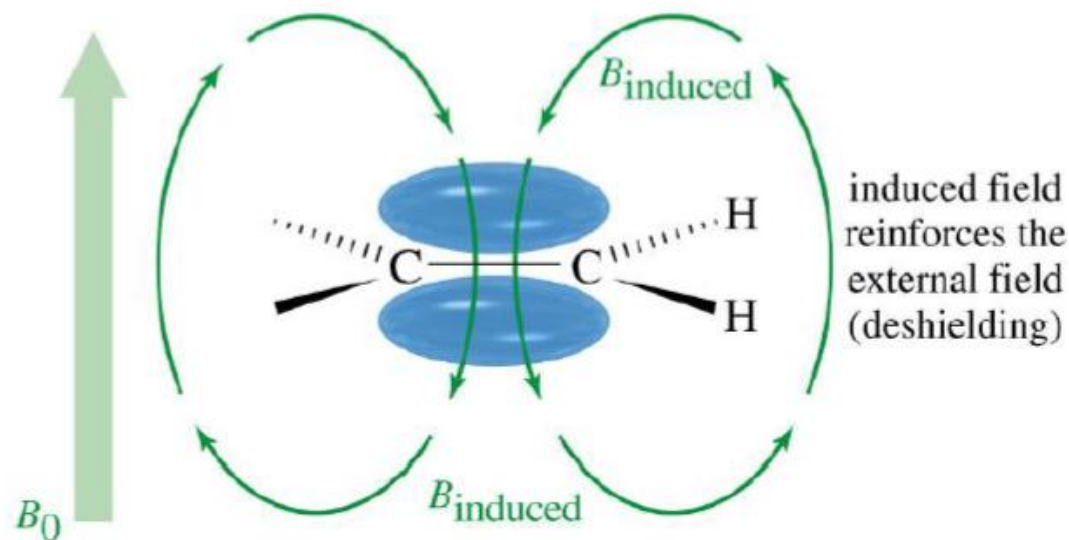
$\delta = 9-10 \text{ ppm}$

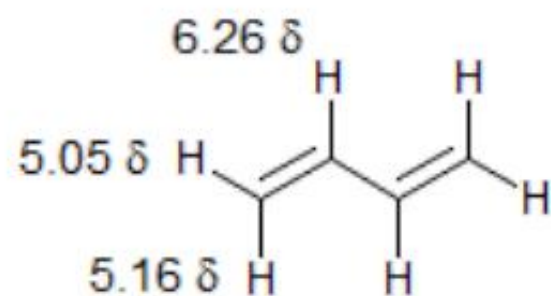
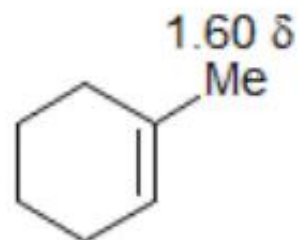
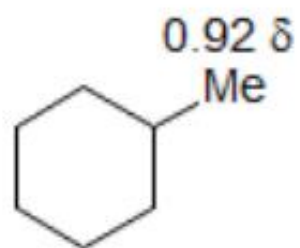
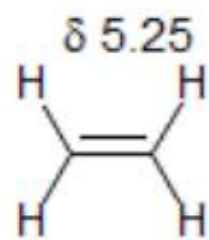
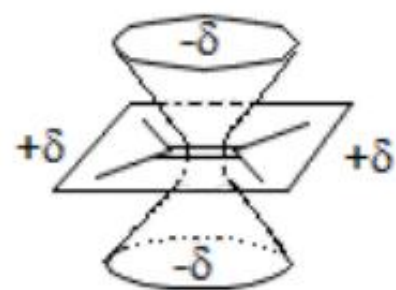


$\delta = 2-3 \text{ ppm}$

Alkene Protons

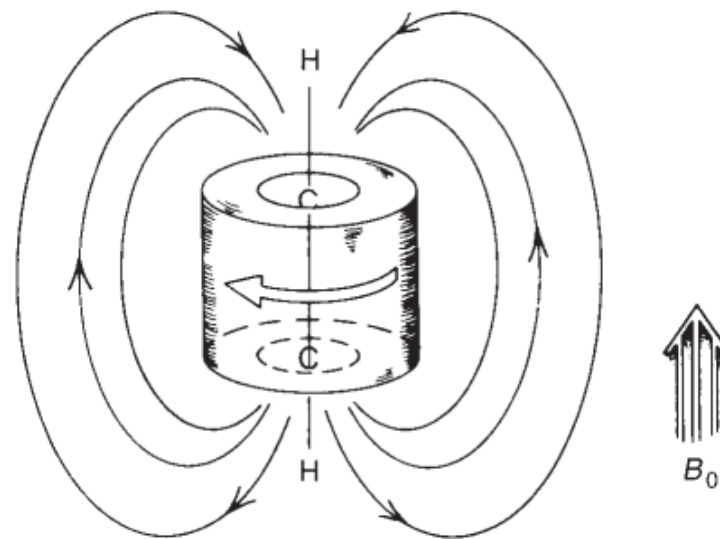
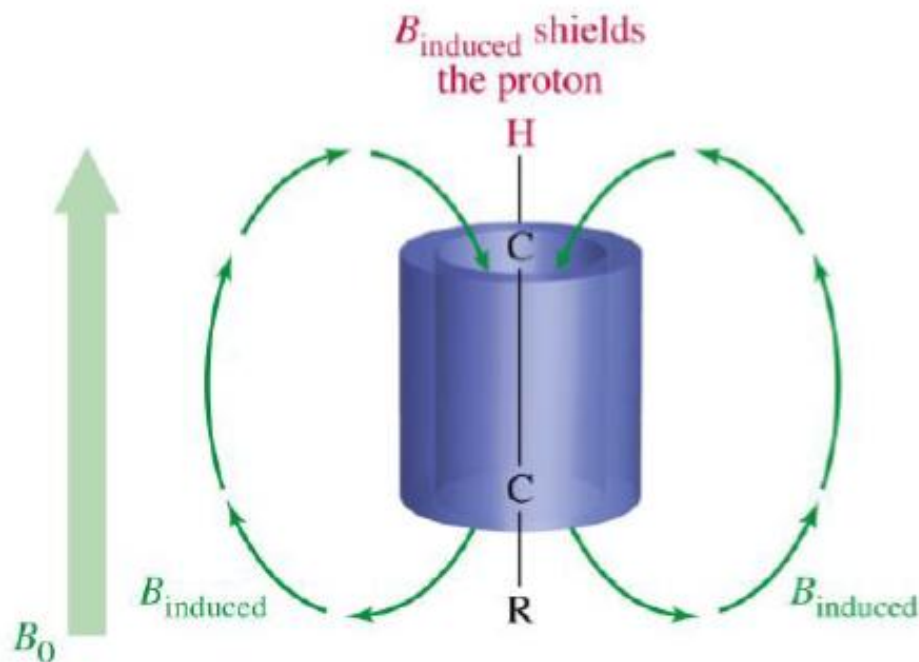
An alkene molecule in an external magnetic field is so oriented that the plane of the double bond is perpendicular to the applied magnetic field. Thus, the circulation of π -electrons is perpendicular to the carbon-hydrogen framework of the molecule. As a result the induced magnetic field caused by the circulation of π -electrons is in the opposite direction of applied magnetic field (diamagnetic) around the carbon atoms and in the direction of applied magnetic field (paramagnetic) in the region of olefinic protons. Therefore, olefinic protons experience greater field strength and consequently resonate at larger value of chemical shift.



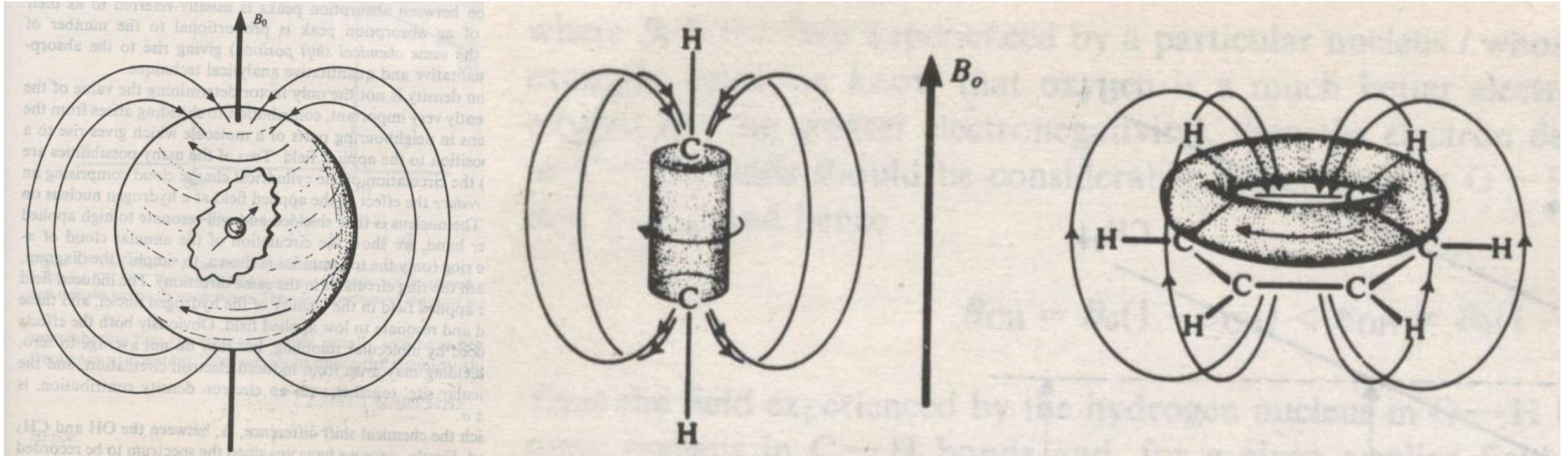


(ii) Alkyne protons

An alkyne molecule when placed in an external magnetic field is so oriented that the plane of the triple bond lies parallel to the direction of applied magnetic field. The induced magnetic field caused by the circulation of π -electrons of the triple bond is in the opposite direction of applied field (diamagnetic) in the region of acetylenic protons. Therefore acetylenic protons are shielded and experience lesser field strength and consequently resonate at smaller value of chemical shift.



Q. So acetylenic protons experience shielding effect whereas aromatic protons feel deshielding effect. Why such difference?

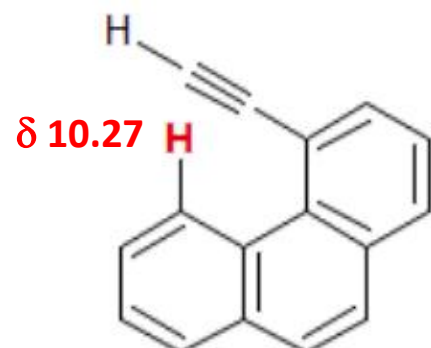
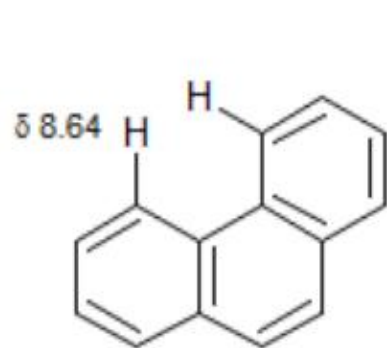
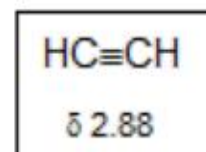
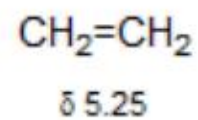
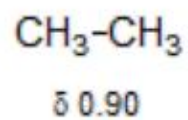
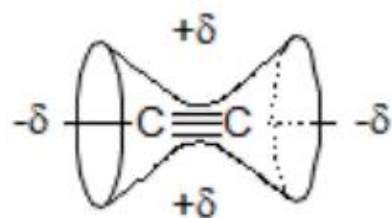


Diamagnetic anisotropy in acetylene

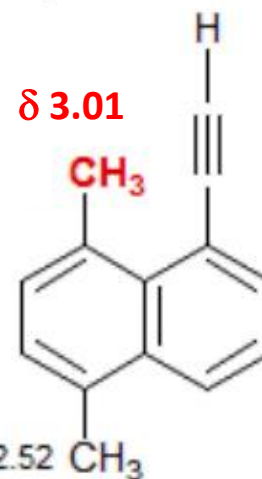
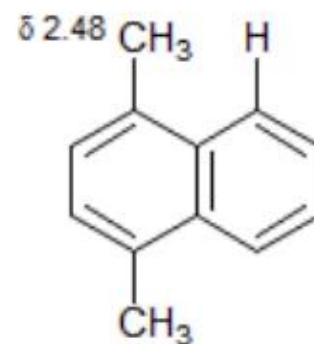
- Linear molecule; symmetrical π -electron cloud around the bond
- When aligned with the field, the π -electrons circulate at right angles
- Induced magnetic field opposes the applied field
- The protons along the bond axis are shielded
- The molecule may not be aligned with the field All orientations are possible
- When perpendicular to the field, the effect is opposite and weak as the rotation of electrons is restricted, the observed chemical shift is an average of all the possible orientations

Diamagnetic anisotropy in ethylene

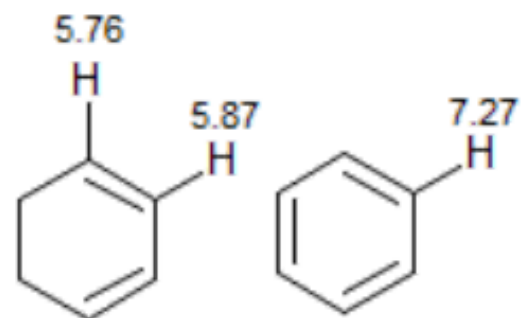
- π -electron cloud above and below the double bond
- When aligned perpendicular to the field, the π -electrons circulate at right angles
- Induced magnetic field augments the applied field at the protons
- The protons are deshielded (observed at low field)



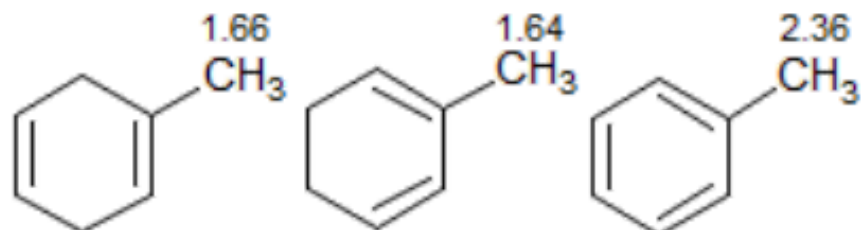
$\Delta\delta$ 1.63 ppm
JOC 1984, 1323



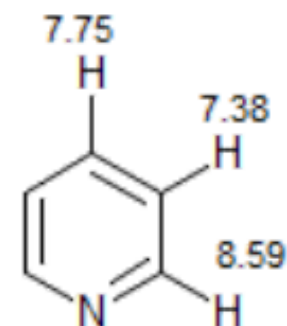
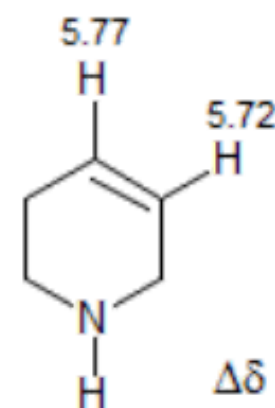
$\Delta\delta$ 0.49 ppm
JOC 1984, 1323



$\Delta\delta = 1.40$ ppm



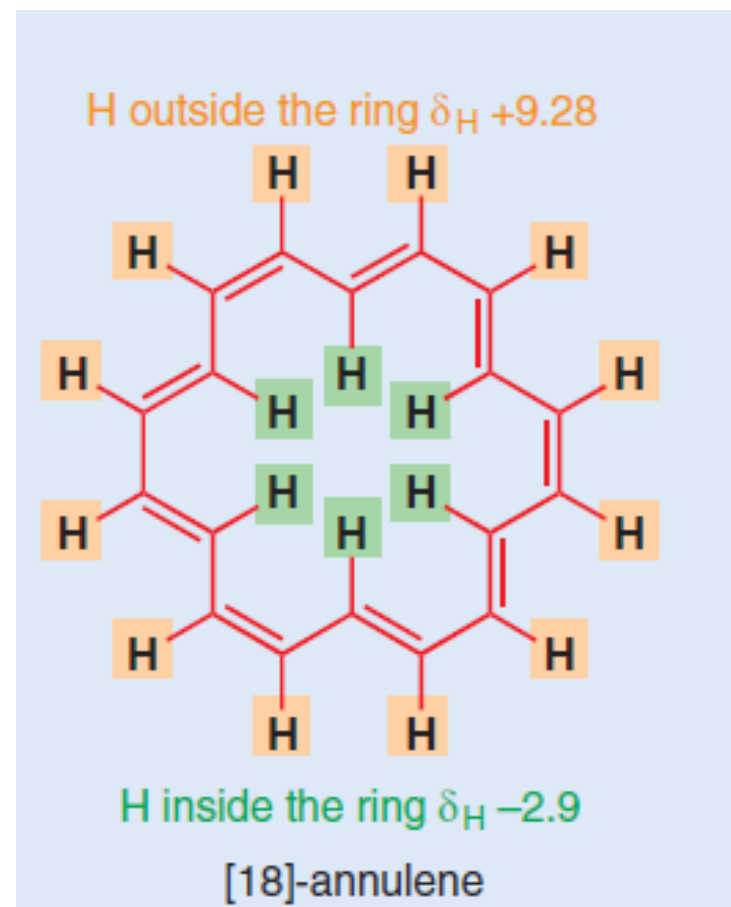
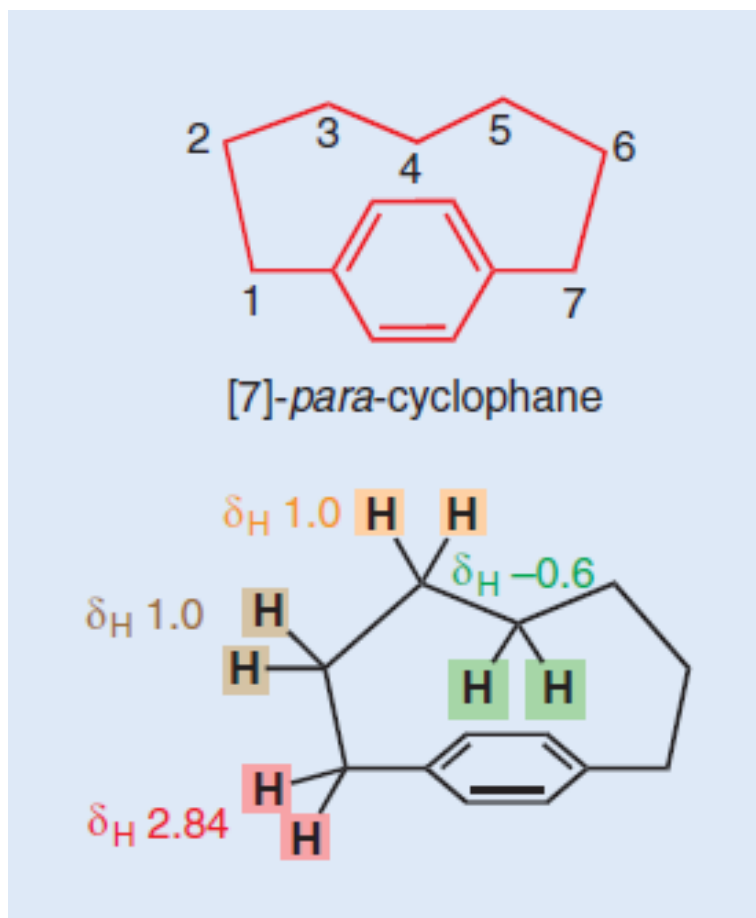
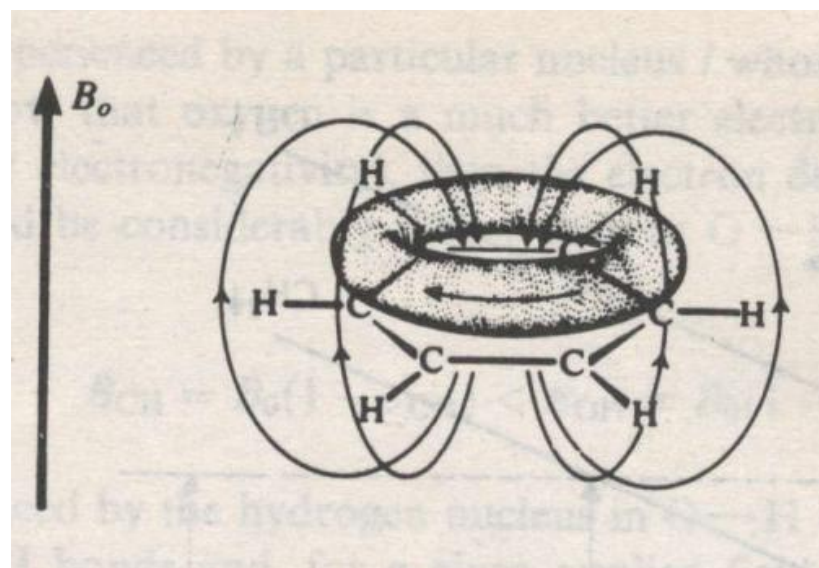
$\Delta\delta = 0.72$ ppm



$\Delta\delta = 1.66$ ppm

Cyclophanes and annulenes:

Cyclophanes have loops of saturated carbon atoms attached at both ends to the same benzene rings. The four H atoms on the benzene ring itself appear as one signal at 7.07 ppm—a typical ring-current deshielded value for a benzene ring. The two CH_2 groups joined to the benzene ring (C_1) are also deshielded by the ring current at 2.64 ppm. But the middle CH_2 group in the chain (C_4) must be pointing towards the ring in the middle of the π system and is heavily shielded by the ring current at -0.6 ppm. In [18] annulene, the hydrogens outside the ring resonate in the aromatic region at rather low field (9.28 ppm) but the hydrogen atoms inside the ring resonate at an amazing -2.9 ppm, showing the strong shielding by the ring current.



The alkene region and the benzene region

In ^{13}C NMR, alkene and benzene carbons came in the same region of the spectrum, but in the ^1H NMR spectrum the H atoms attached to arene C and alkene C atoms sort themselves into two groups. Look at the ^{13}C and ^1H chemical shifts of cyclohexene and benzene. The two carbon signals are almost the same (1.3 ppm difference, < 1% of the total 200 ppm scale) but the proton signals are very different (1.6 ppm difference = 16% of the 10 ppm scale).

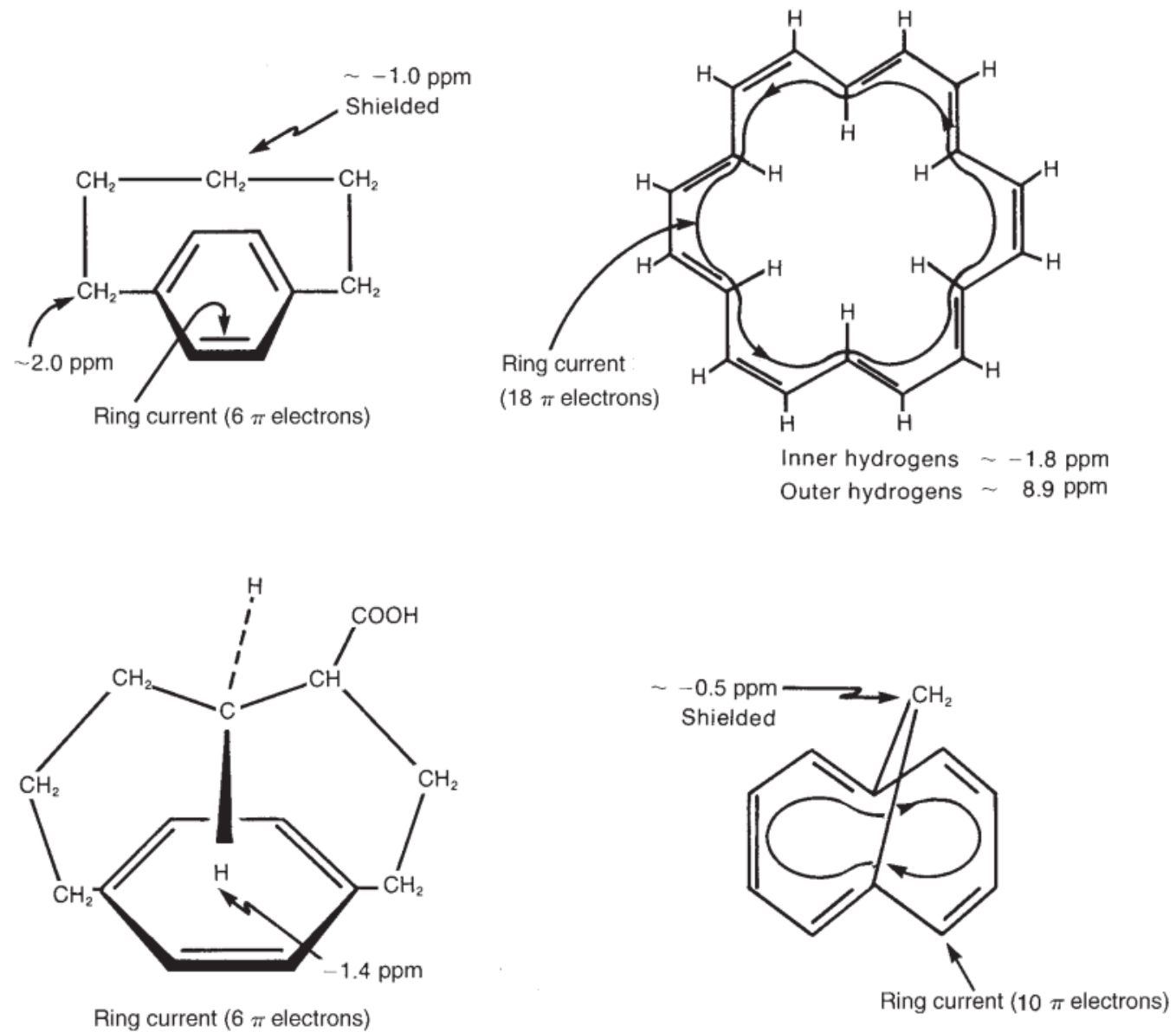
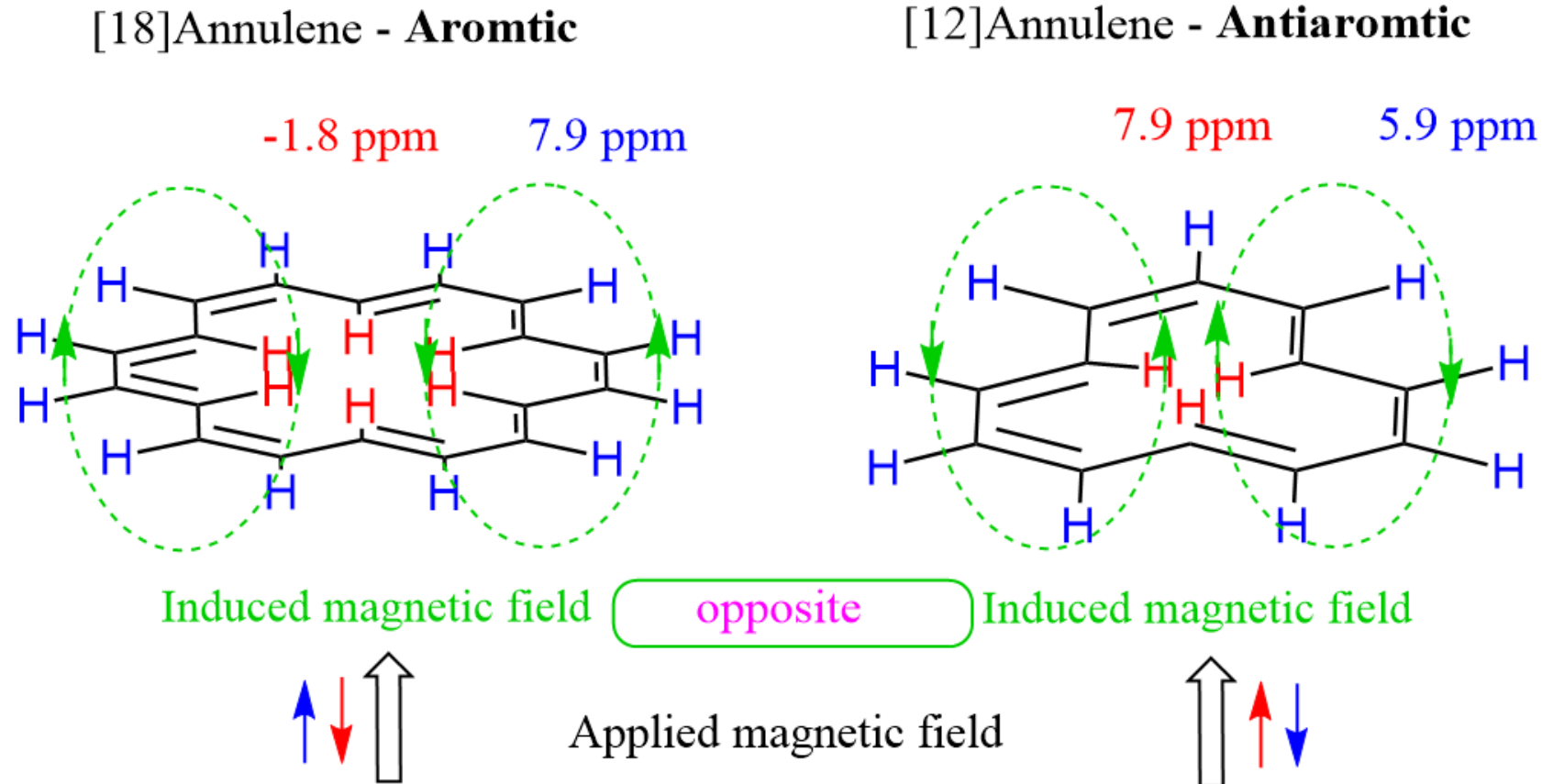


FIGURE 5.24 The effects of anisotropy in some actual molecules.

Aromatic and Antiaromatic - NMR Chemical Shifts



The different ring current generates induced magnetic field with opposite directions in aromatic and antiaromatic compounds.

Aromatic - inner protons appear in lower ppm region.

Antiaromatic - inner protons appear in higher ppm region.